

## LETTER

# Integrating Artificial Intelligence Into Endoscopy Education: Evidence, Gaps, and Future Directions

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We read with great interest the comprehensive review by Ho et al. [1] summarizing artificial intelligence (AI) applications in endoscopy training. The collated randomized and prospective data provide useful benchmarks for educators. Notably, computer-aided detection (CADe) improved overall adenoma detection among trainees (57.5% vs. 44.5%; absolute +13.0%) and increased adenoma detection in beginners (60.0% vs. 41.9%; adjusted relative risk [RR] 1.58, i.e., ~58% relative increase), demonstrating a clear short-term training benefit [2]. Earlier multicenter work similarly showed higher adenoma detection rate (ADR) with CADe (54.8% vs. 40.4%) [3].

We applaud the authors' balanced discussion of advantages and potential harms. To further strengthen the practical roadmap, we suggest two focused additions. First, quantifying durability: multicenter longitudinal audits or scheduled "AI-off" assessments would clarify whether short-term ADR gains translate into sustained independent competency. Second, expanding the section on training metrics and simulation would be valuable. For example, contrast-harmonic endoscopic ultrasound (CH-EUS) AI tools achieved a Dice similarity coefficient of approximately 0.76 and improved trainee intersection over union (IoU) from 0.80 to 0.87 with faster lesion localization, indicating that AI systems can approximate expert trainer guidance for specific segmentation tasks [4]. Similarly, computer-based capsule endoscopy training raised trainee test scores from 49.5% to 62.1% post-training, illustrating that structured computer-based learning (CBL) modules can materially enhance lesion recognition skills [5].

Overall, Ho et al. provide an excellent synthesis of current evidence. Emphasizing measurable long-term outcomes, robust data governance frameworks, and standardized outcome sets

for AI-augmented training would help the field transition from promising short-term trials to sustainable curricula that preserve clinician autonomy while safely leveraging AI.

## Author Contributions

**Weihao Cheng:** formal analysis, writing – original draft, writing – review and editing. **Shangxuan Li:** formal analysis, writing – review and editing. **Zekai Yu:** investigation, formal analysis, writing – review and editing. The author has read and agreed to the published version of the manuscript.

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The authors have nothing to report.

## Conflicts of Interest

The authors declare no conflicts of interest.

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